SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	1 july2026
Team ID	xxxxxx
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML5, CSS3, Bootstrap, JavaScript	*Used to create a responsive, user-friendly web interface where users can enter soil and environmental parameters and view crop prediction results.*

2	**Backend / Server-Side**	*Python, Flask*	*Flask provides a lightweight web framework to handle user requests, process input data, load the machine learning model, and return crop recommendations efficiently.*
3	**Database / Data Storage**	*CSV Dataset (Kaggle Crop Recommendation Dataset)*	e*The application uses a preprocessed CSV dataset for training the machine learning model. No separate database is required because predictions are generated using the trained model.*
4	**Cloud / Hosting / Deployment**	Render	*Render is used to deploy and host the Flask application online, making it accessible from anywhere through a public URL.*
5	**Version Control & CI/CD**	Git, GitHub	*Git is used for version control, while GitHub stores the project repository, tracks changes, and integrates with Render for automatic deployment after code updates.*

6	**Third-Party APIs / Other Tools**	*Scikit-learn, Pandas, NumPy, Joblib, Gunicorn*	Scikit-learn is used to train the crop recommendation model. Pandas and NumPy handle data processing. Joblib loads the trained model efficiently, and Gunicorn serves the Flask application in production on Render.